

JupyterLite

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Intro.ipynb

Notebook Python (Pyodide)

```
[1]: import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
[2]: from sklearn.datasets import load_breast_cancer

data = load_breast_cancer()
X = data.data
y = data.target

print("Shape of X:", X.shape)
print("Shape of y:", y.shape)
```

Shape of X: (569, 30)
Shape of y: (569,)

```
[3]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
[4]: model = LogisticRegression(max_iter=500)
model.fit(X_train, y_train)
```

/lib/python3.12/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

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Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(

[4]: LogisticRegression
LogisticRegression(max_iter=500)

[5]: y_pred = model.predict(X_test)

[6]: print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

Accuracy: 0.956140350877193

Confusion Matrix:
[[39 4]
 [1 70]]

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.91	0.94	43
1	0.95	0.99	0.97	71
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114

[]:

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[4]: model = LogisticRegression(max_iter=500)
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/lib/python3.12/site-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
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Classification Report:
precision    recall  f1-score   support


```

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

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